

Synogize New Source Delivery Handoff

Pentana Vehicle Sales Cleaning, Silver Split, Gold Schema, and Marts

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1 Summary of Key Assumptions

This handoff report documents the current delivery state for the new source table PENTANA_DB.RAW.VEHICLE_SALES_NEW. The work now includes a cleaned Silver core table, a split Silver layer by business domain, a Gold star schema, and starter analytical mart views.

1.1 Main modelling assumptions

- VIN is the practical vehicle-level primary key for the new source and is the main linkage key across the split Silver tables.
- VIN is kept even when it is not a standard 17-character VIN. Format issues are flagged rather than excluded.
- VIN duplicates and exact duplicate rows are not treated in this final version because they were removed by William for separate analysis later.
- DELIVERYDATE is the strongest event date for the Gold fact linkage.
- Delivery geography is preferred in the fact table, with fallback to dealer geography when delivery postcode is missing.

1.2 Cleaning assumptions

- Valid working date range: 2010-01-01 to 2030-12-31.
- DELIVERYDATE is the most trusted core event date.
- ESTIMATEDARRIVALDATE, ESTIMATEDDELIVERYDATE, and SCHEDULEDELIVERYDATE are treated as planning dates, not confirmed event dates.
- COMPLIANCEPLATEDATE is treated more cautiously because it can appear in non-standard formats.
- Date chronology conflicts are flagged for monitoring, including order, deal, invoice, delivery, arrival, and requested or customer delivery sequence issues.
- Core commercial thresholds reuse the earlier P99-based working ranges.
- SALEAMOUNT, VEHICLESALEPRICE, and VEHICLECOSTPRICE: 0 to 1,000,000.
- DISCOUNTAMOUNT and OTHERDISCOUNTAMOUNT: 0 to 250,000.
- PAINT* and TRIM*: 0 to 250,000.
- DELIVERY*, REGISTRATION*, and CTP*: 0 to 100,000.
- STAMPDUTY* and FLEETCLAIMAMOUNT: 0 to 250,000.
- Total amount fields: 0 to 2,000,000; TOTALDEPOSITAMOUNT: 0 to 1,000,000; trade-in and customer totals: 0 to 2,000,000.
- Profit fields: -1,000,000 to 1,000,000; negative profit values are allowed and not nulled just for being below zero.
- ODOMETER is accepted only from 0 to 500,000.

- `ENGINE_SIZE_IN_CC` is accepted only from 500 to 8,000.
- `VEHICLESALEPRICE < VEHICLECOSTPRICE` is flagged, not nulled.
- Categorical cleaning uses trimmed and normalised text, mainly through whitespace cleanup and uppercase standardisation before business rules are applied.
- For business categoricals, values that clearly do not make sense for the field are removed from the cleaned version and kept only as null or flagged outcomes.
- `SALETYPE` and `SALETYPE_2` are both retained as cleaned fields; each is trimmed and normalised, and `SALETYPE_FINAL` is only populated when the combined result is unambiguous.
- Category values such as `USED VEHICLE` were not accepted as valid sale-type values in the cleaned layer.
- `WHOLESALEFLAGYN` is treated as a binary field, and date-like values are invalid.
- Dealer and factory option placeholder values such as `0`, `0.00`, `BASE`, and `ACCESSORY NOT ON FILE` are treated as null in cleaned option fields.

1.3 Current validation status

- Silver core row count matches raw source: 5,075,776.
- `VIN_CLEAN` is non-null and unique in the Silver core.
- Split Silver tables were built successfully.
- Gold dimensions, fact table, and starter marts were built successfully.
- `DIM_STATE` was reduced to a cleaner grouped state set with 10 rows.

2 Instructions for the Three Main Build Files

The full new-source pipeline can be rebuilt from three main files. Each file also has a matching SQL version for direct Snowflake worksheet execution.

2.1 File 1: Bronze and Silver core creation

Python: `scripts/new_source/11_bronze_silver_creation_new.py`

SQL: `sql/new_source/11_bronze_silver_creation_new.sql`

This step:

- creates `BRONZE_NEW`, `SILVER_NEW`, and `GOLD_NEW` if needed
- reads from the raw source table
- builds the cleaned Silver core table `VEHICLE_SALES_SILVER_CORE_T`

2.2 File 2: Silver split

Python: scripts/new_source/12_silver_split_new.py

SQL: sql/new_source/12_silver_split_new.sql

This step:

- reads from `VEHICLE_SALES_SILVER_CORE_T`
- creates domain-specific Silver tables for vehicle details, geography, sales process, dealer options, factory options, and finance support
- keeps VIN as the main linkage key across the split tables

2.3 File 3: Gold schema and marts

Python: scripts/new_source/13_gold_schema_new.py

SQL: sql/new_source/13_gold_schema_new.sql

This step:

- builds the four required dimensions
- builds the central fact table
- builds a first set of mart views for business analysis

2.4 How to run the files

- Local Python mode: use the Python files and the local Snowflake connection settings.
- Snowflake Python mode: use the same Python files with an existing `session` object.
- Snowflake worksheet mode: use the SQL files directly.

3 Built Tables and What They Contain

3.1 Silver core table

`PENTANA_ANALYTICS.SILVER_NEW.VEHICLE_SALES_SILVER_CORE_T`

This is the main cleaned row-grain transaction table. It contains:

- cleaned identifiers such as `VIN_CLEAN`, `DEALID_CLEAN`, `INVOICEID_CLEAN`, and `REGO_FINAL`
- cleaned dates and date issue buckets
- chronology flags
- cleaned sales, cost, discount, profit, deposit, and trade-in amounts
- cleaned sale-type and wholesale logic

- core vehicle cleaning outputs such as SEATS_FINAL, MODEL_YEAR_FINAL, SERIES_FINAL, ODOMETER_CLEAN, and ENGINE_SIZE_IN_CC_CLEAN

3.2 Split Silver tables

Table	Purpose
VEHICLE_DETAILS_SILVER_T	One row per VIN. Stable vehicle descriptors used as the parent for DIM_VEHICLE.
VEHICLE_GEOGRAPHY_SILVER_T	Cleaned postcode, state, country, and state-fill logic from postcode.
VEHICLE_SALES_PROCESS_SILVER_T	Cleaned dates, identifiers, chronology flags, sale type logic, and process-oriented fields.
VEHICLE_OPTIONS_DEALER_SILVER_T	Dealer-added option identifiers, descriptions, and amounts.
VEHICLE_OPTIONS_FACTORY_SILVER_T	Factory-added option identifiers, descriptions, and amounts.
VEHICLE_FINANCE_SUPPORT_SILVER_T	Finance company, finance type, finance contract, and finance amounts.

All split Silver tables currently keep row-level lineage through VIN, and when needed also preserve DEALID_CLEAN and INVOICEID_CLEAN.

4 Gold Dimensions and Fact Table

4.1 Dimensions

Table	Content
DIM_VEHICLE	Uses VIN as the natural key. Includes make, model year, series, body style, fuel type, transmission type, seats, vehicle type, new or used, carline, model description, drive description, paint description, and trim description.
DIM_DATE	Calendar dimension generated from cleaned order, deal, invoice, and delivery dates. Includes date key, calendar date, year, quarter, month, and weekday attributes.
DIM_STATE	Cleaned grouped state dimension. Current groups are Australian states, NZ, and OTHER.
DIM_POSTCODE	Cleaned postcode dimension using dealer and delivery postcodes, plus the linked state code and postcode source.

4.2 Fact table

FACT_VEHICLE_SALES is the central Gold fact table.

It currently includes:

- dimensional linkage keys:
 - VEHICLE_KEY
 - DATE_KEY
 - STATE_KEY
 - POSTCODE_KEY
- sales measures such as SALEAMOUNT, VEHICLESALEPRICE, and finance or delivery-related sale amounts
- cost measures such as VEHICLECOSTPRICE, delivery cost, registration cost, CTP cost, and total option or other charge costs
- profit and discount measures such as DISCOUNTAMOUNT, OTHERDISCOUNTAMOUNT, and gross profit fields
- customer and cash measures such as deposit, trade-in allowance, trade-in net, and total customer amount

4.3 Current Gold validation highlights

- DIM_VEHICLE: 5,075,776 rows
- DIM_DATE: 4,390 rows
- DIM_STATE: 10 rows
- DIM_POSTCODE: 5,174 rows
- FACT_VEHICLE_SALES: 5,075,776 rows

5 Current Mart Views and Analysis Opportunities

5.1 Built mart views

View	Purpose
MART_VEHICLE_SALES_OVERVIEW_V	Monthly overview by make using the fact table plus the date and vehicle dimensions.
MART_MAKE_STATE_SALES_V	Sales and customer amount view by make and cleaned state.
MART_FINANCE_OPTION_SUMMARY_V	Focused view of finance-related sales and option-related sales.
MART_DEALER_POSTCODE_MAKE_SALES_V	Dealership-oriented view using dealer post-code and make. Useful for dealership performance and brand concentration analysis.
MART_MARGIN_ANALYSIS_V	Margin-oriented view using sale price, cost price, discount amounts, and derived margin calculations.

5.2 Two especially useful views

Dealer postcode and make `MART_DEALER_POSTCODE_MAKE_SALES_V` is useful for analysing:

- which makes are strongest by dealer location
- postcode-level dealership concentration
- revenue and gross profit by make and dealer postcode

Margin analysis `MART_MARGIN_ANALYSIS_V` is useful for:

- comparing vehicle sale price against vehicle cost price
- looking at margin pressure from discounts
- analysing margin by make, state, period, and new versus used

5.3 Recommended next analytical additions

- margin by dealer postcode and make
- margin by fuel type and new versus used
- finance penetration by make and state
- dealer options versus factory options by make
- delivery postcode versus dealer postcode movement analysis